PRDA–03

HR ANALYTICS

Workforce Analytics & Insights Report

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Project: PRDA-03 – HR Analytics

Prepared using Microsoft Power BI

1. EXECUTIVE SUMMARY

This report presents a descriptive analysis of the HR workforce using the Power BI dashboard prepared for PRDA–03. The analysis focuses on workforce size, work arrangement, age profile, employee distribution by hire year, gender composition, geographic distribution, departmental structure and race distribution.

The dashboard indicates a workforce of 22,214 employees with an average age of 38.43 years. Remote employees account for 75.25% of the workforce, while 24.75% are based at headquarters. The gender distribution is relatively balanced between male and female employees, with non-conforming employees representing a smaller share.

OVERALL ANALYTICAL TAKEAWAY

The strongest structural pattern is the concentration of employees by work arrangement and geography: remote work is the dominant arrangement, while Ohio accounts for a very large share of the workforce compared with the other states shown.

2. DATA & METHODOLOGY

The analysis was developed in Microsoft Power BI using the cleaned HR dataset embedded in the submitted PBIX file. The dashboard uses employee-level records and aggregates them into the requested HR measures and dimensions.

Analysis area	Measure / dimension	Purpose
Workforce size	Employee ID count	Measure total employee population
Work arrangement	Headquarters vs Remote	Measure location mix
Age	Average Age	Summarize workforce age profile
Hiring	Employee ID count by hire year	Show employee distribution by hire year
Gender	Gender	Show workforce gender composition
Geography	Location State	Compare employee concentration by state
Department	Department	Compare workforce allocation across functions
Race	Race	Show demographic composition

CALCULATION NOTE:

The visual titled “Hiring Trend by Year” uses Count of Employee ID grouped by Hire Year. Because the dataset represents employee records, this visual shows the distribution of current employee records by hire year rather than confirmed historical hiring volume. It should therefore be interpreted as a hire-year cohort distribution, not as a historical recruitment-rate measure.

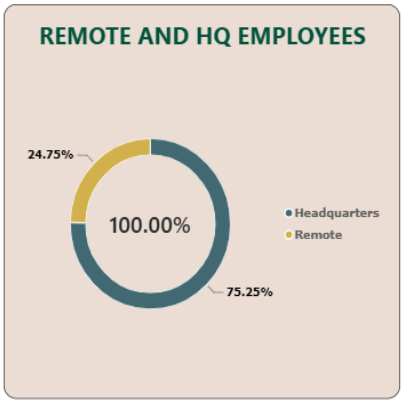
3. KEY PERFORMANCE INDICATORS

Total Employees	Remote Workforce	Average Employee Age
22,214	75.25%	38.43 years

The work-location split shown in the dashboard is 75.25% Remote and 24.75% Headquarters, giving a combined 100% of the displayed workforce.

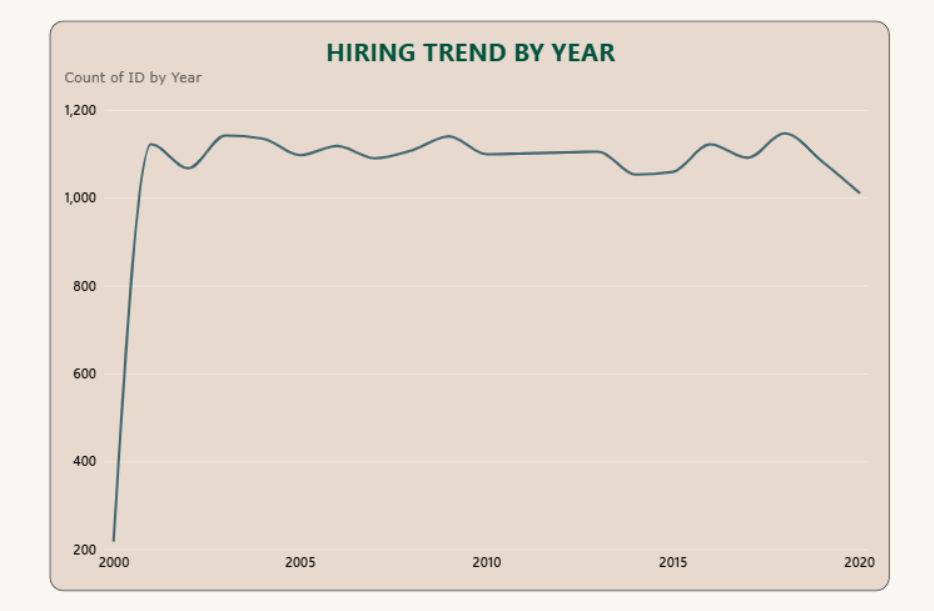
- The workforce size is 22,214 employees based on the dashboard KPI.
- The average employee age is 38.43 years.
- Remote work is the dominant work arrangement at 75.25%, compared with 24.75% at headquarters.







4.HIRING TREND BY YEAR

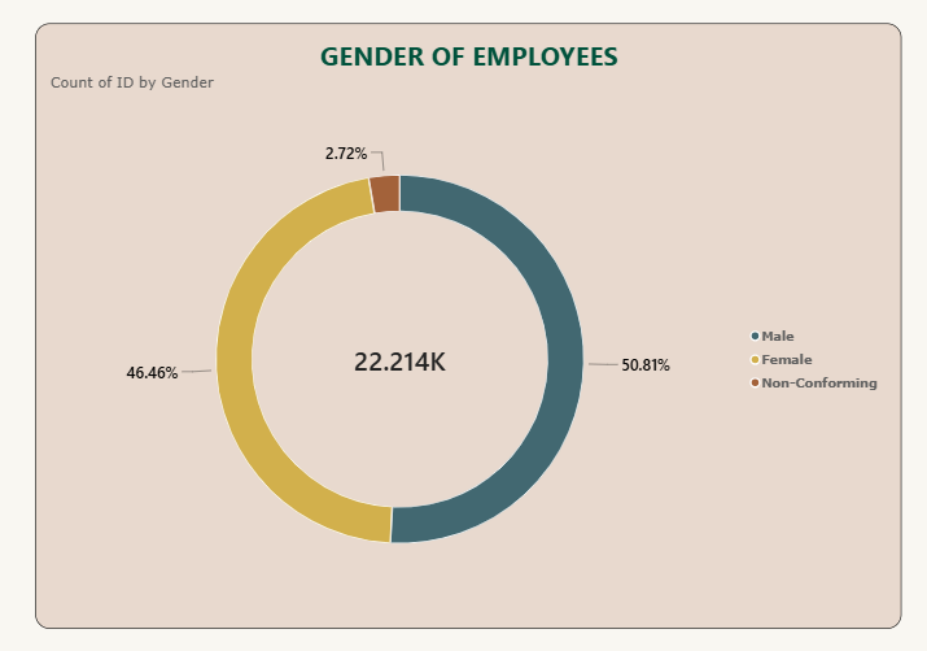


The number of current employee records associated with each hire year rises sharply from the earliest displayed year into the early 2000s, after which it remains broadly within the 1,000–1,150 range for much of the series, with moderate fluctuations in later years.

The series reaches a relatively high point around 2018 before declining toward 2020. Overall, the distribution of current employees across hire years remains relatively stable across much of the displayed period, with a noticeable reduction at the end of the series.

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5. GENDER DISTRIBUTION



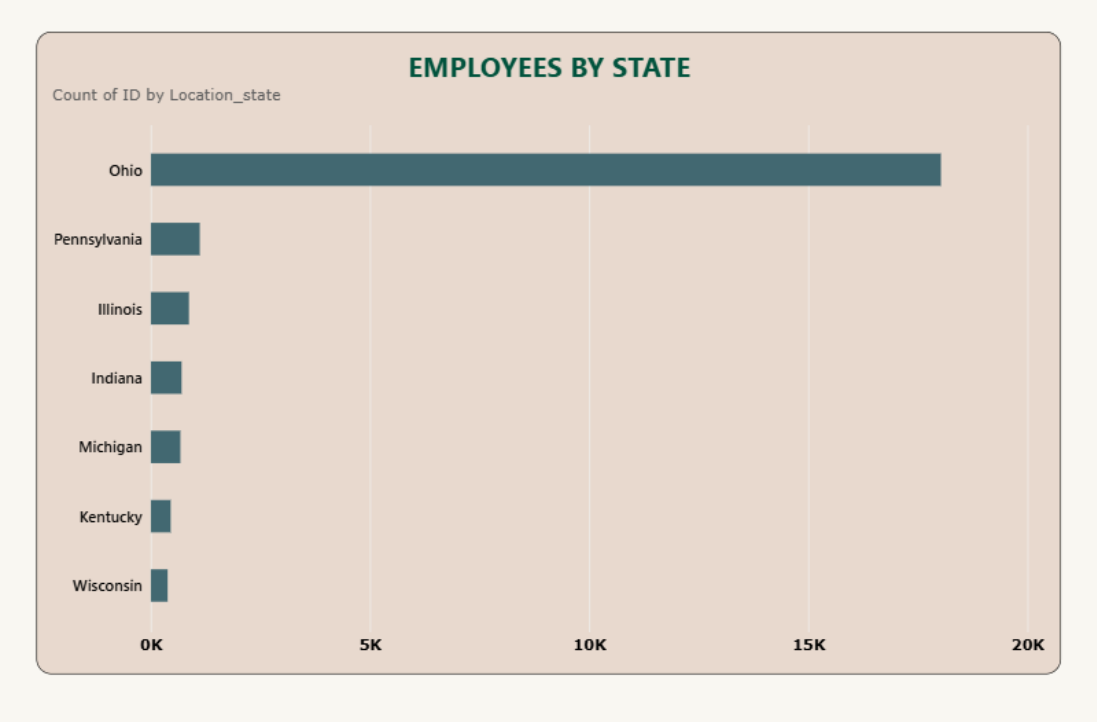
The workforce is distributed across three gender categories shown in the dashboard: Male (50.81%), Female (46.46%) and Non-Conforming (2.72%).

Male employees form the largest category, but the difference between the male and female shares is relatively small at approximately 4.35 percentage points. The non-conforming category represents a smaller but visible share of the workforce.

ANALYTICAL IMPLICATION

Gender composition should be monitored over time rather than evaluated from a single snapshot. Trend-based monitoring can show whether workforce representation is changing across departments, locations and hiring cohorts.

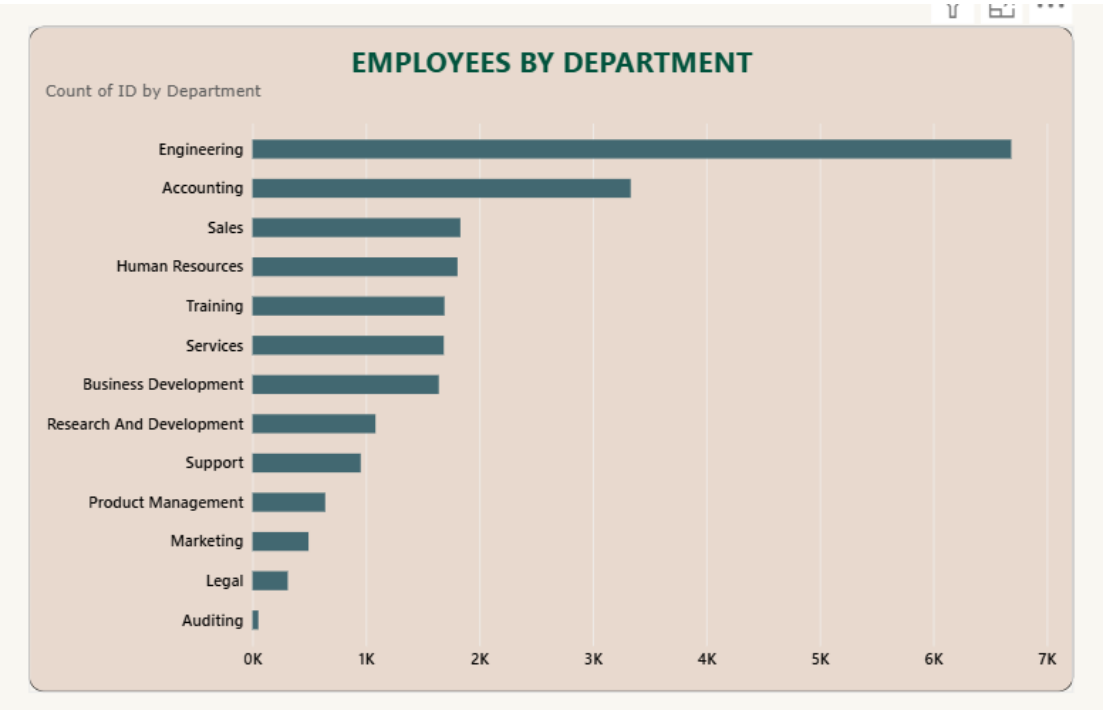
6. EMPLOYEES BY STATE



The state distribution is highly concentrated. Ohio is the dominant location in the displayed data, with a substantially larger employee population than Pennsylvania, Illinois, Indiana, Michigan, Kentucky and Wisconsin.

This concentration means that changes in the Ohio employee population could have a comparatively large effect on the geographic profile of the states displayed. The smaller state populations may also be relevant when evaluating regional recruitment and workforce allocation.

7. EMPLOYEES BY DEPARTMENT



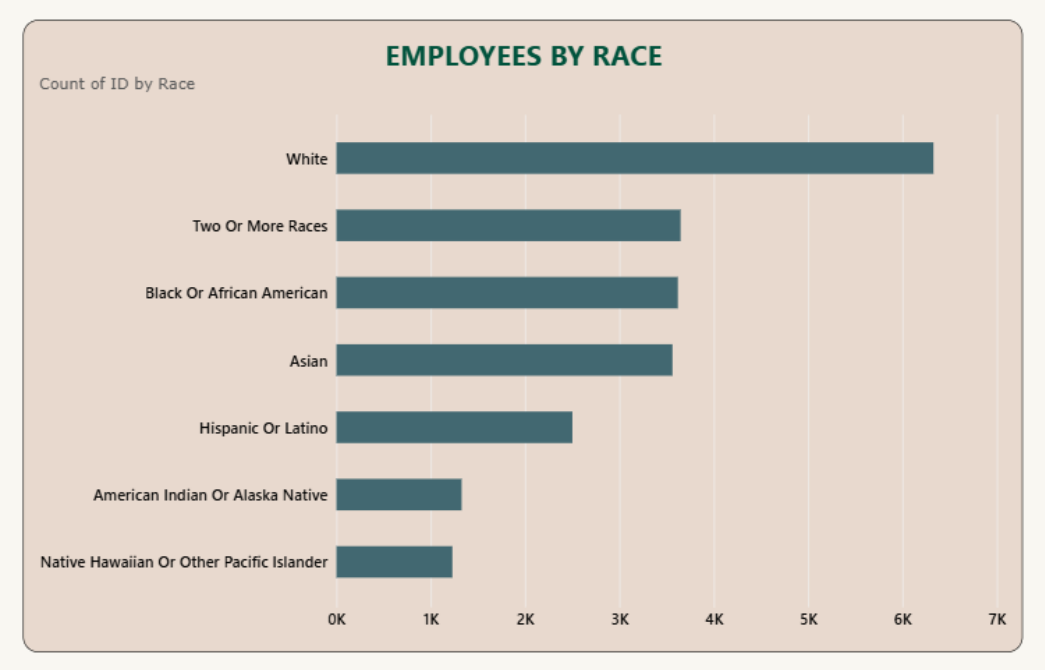
Engineering is the largest department in the dashboard by a clear margin, followed by accounting. Sales, Human Resources, Training, Services and Business Development form a second group of medium-sized functions, while Research and Development, Support, Product Management, Marketing, Legal and Auditing have smaller workforce counts.

The variation in department headcount supports using department-level staffing measures rather than relying only on total employee count for workforce planning. Engineering has the largest employee population, while several other departments have substantially smaller headcounts.

PLANNING IMPLICATION

Department-level headcount can be combined with actual recruitment history, attrition and workload measures in future analysis to identify where staffing demand is changing.

8. EMPLOYEES BY RACE



White employees form the largest racial category in the displayed workforce. Two or More Races, Black or African American and Asian categories form the next group, followed by Hispanic or Latino, American Indian or Alaska Native, and Native Hawaiian or Other Pacific Islander.

The chart demonstrates that the workforce contains multiple racial groups with different levels of representation. Because these are aggregate counts, the visualization describes composition but does not by itself explain the causes of those differences.

9. INTEGRATED INSIGHTS

1. Remote work is structurally significant:

At 75.25%, remote employees represent three times the share of headquarters employees (24.75%). This makes remote-work capacity, collaboration practices and workforce-support processes material considerations for HR planning.

2. Geographic concentration is pronounced:

Ohio contains the largest share of employees among the states displayed in the dashboard, while the remaining displayed states have much smaller populations.

3. Engineering is the largest function:

Engineering has the largest departmental headcount by a substantial margin, with accounting second. The remaining departments are distributed across a wider range of smaller workforce sizes.

4. Employee distribution by hire year is broadly stable after the initial increase:

Apart from the sharp increase at the beginning of the series, employee counts associated with individual hire years mostly remain around the 1,000–1,150 level, with moderate year-to-year fluctuations and a noticeable decline toward 2020. This pattern describes the distribution of current employees across hire years rather than confirmed historical hiring volume.

5. Gender composition is relatively close between the two largest categories:

Male employees account for 50.81% and female employees account for 46.46%, while non-conforming employees account for 2.72%. The two largest groups therefore have similar representation, with a difference of 4.35 percentage points.

6. The workforce has a multi-group demographic profile:

The race chart shows meaningful representation across several categories. This is useful as a baseline for monitoring workforce composition over time.

7. Average age provides a useful baseline:

The average age of 38.43 years can serve as a baseline metric when combined with age bands, tenure and retirement/attrition data in a future workforce-planning model.

10. COMPANY RECOMMENDATIONS

Workforce planning — Use the employee distribution by hire year as a descriptive baseline and combine it with actual recruitment, attrition, tenure and open-position data before setting future hiring targets.

Remote-work operations — Because 75.25% of employees are remote, review collaboration, onboarding, communication, cybersecurity and remote employee support processes alongside headquarters capacity.

Geographic planning — Because Ohio dominates the displayed state distribution, evaluate location concentration alongside business requirements, workforce demand and operational needs before considering changes to regional workforce allocation.

Department capacity — Track headcount by department alongside workload, productivity and attrition measures. The large Engineering workforce should be evaluated separately from smaller functions rather than applying the same staffing assumptions across all teams.

Demographic monitoring — Continue tracking gender and race representation over time and, where appropriate, segment the measures by department, location and hiring cohort to understand changes more precisely.

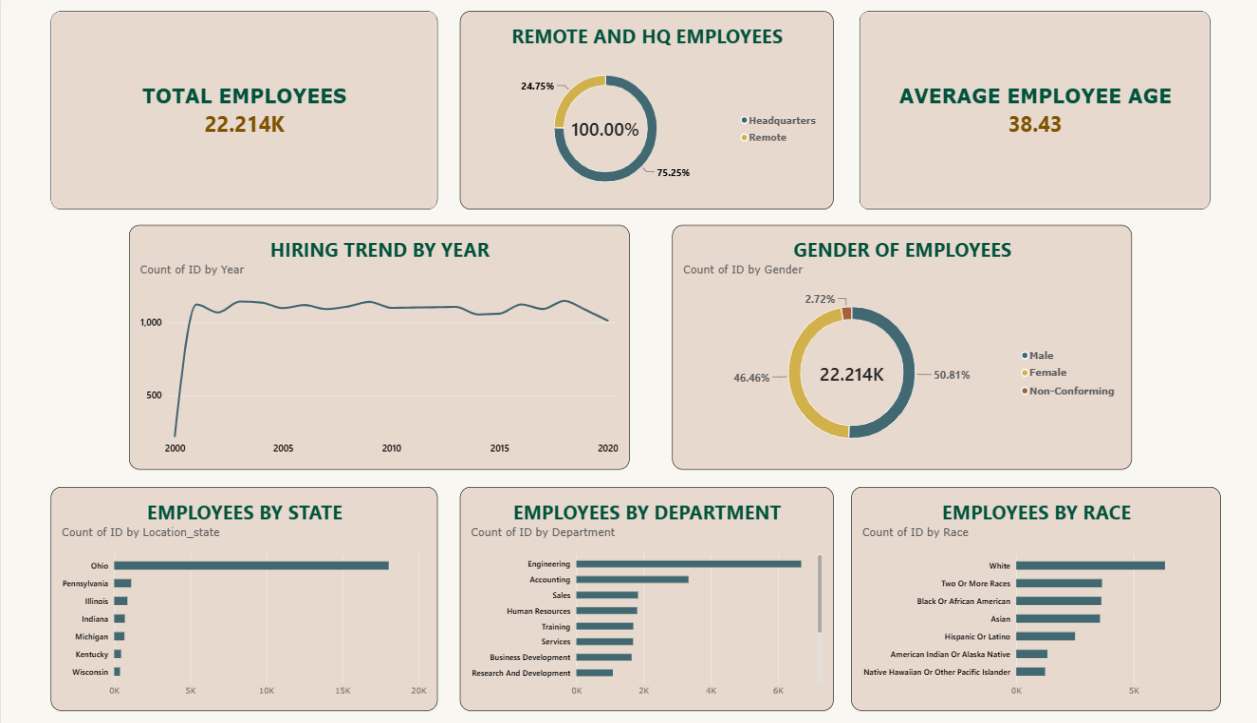
Workforce age and succession — Use the 38.43-year average as a baseline, but supplement it with age bands, tenure and retirement eligibility before making succession-planning decisions.

PRIORITY FOR FUTURE ANALYSIS

The current dashboard is descriptive. Adding attrition, tenure, salary, performance, recruitment-source and open-position data would provide the inputs needed for more advanced workforce planning and potentially predictive analysis.

11. POWER BI DASHBOARD EVIDENCE

The following pages reproduce the submitted Power BI visuals as evidence of the completed analysis.



KEY INSIGHTS

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* The organization has approximately 22.214K employees.
* The average employee age is 38.43 years.
* Remote employees represent 75.25% of the workforce, while Headquarters employees represent 24.75%.
* Male employees represent 50.81% of the workforce, Female employees represent 46.46%, and Non-Conforming employees represent 2.72%.
* The number of employees hired varies across years, showing changes in the hiring trend over time.
* Employee distribution varies across departments, indicating differences in workforce size between departments.
* Employee concentration differs across states, showing variation in workforce distribution by location.
* The workforce includes employees from multiple racial groups, with different levels of representation.

COMPANY SUGGESTIONS

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* Use the year-wise hiring trend to support future workforce planning and recruitment decisions.
* Review employee distribution across departments to identify areas that may require additional workforce support.
* Use state-wise employee distribution to improve workforce allocation and location planning.
* Monitor Remote and Headquarters workforce distribution when planning workplace capacity and hybrid-work policies.
* Continue tracking gender and race representation to support workforce diversity monitoring.
* Use the average employee age along with hiring trends to support long-term workforce planning and succession planning.

CONCLUSION

This HR analytics report provides a descriptive overview of the workforce using the Power BI dashboard. The analysis highlights a workforce of 22,214 employees, a predominantly remote work arrangement, strong geographic concentration in Ohio, and a larger Engineering workforce compared with other departments. The employee distribution by hire year provides a descriptive view of current employees across hire-year cohorts rather than confirmed historical hiring volume. Overall, the dashboard establishes a useful baseline for workforce monitoring. Future analysis incorporating attrition, tenure, salary, performance, recruitment-source and open-position data would support deeper workforce planning and potentially predictive analysis.